G11-02-GATE-A-TOPOLOGY-AUDIT-v1: Vorticity topology - does a well-defined object survive $\nu\Delta\omega$?

ZFL-CX

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$$\boxed{G7 = FROZEN,\ G8 = CLOSED,\ G9\ BLOCKED,\ G10\ CLOSED,\ G11\text{-}01\ CLOSED}$$

$$\boxed{G11\text{-}02 = vorticity\ topology\ pre\text{-}audit,\ GATE\ A\ first}$$

$$\boxed{\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}, C_F^{conc}, C_I^{freq}, C_J^{neg}\}}$$

1 Question A - single mission

$$\boxed{T^* < \infty \overset{?}{\Longrightarrow} \text{topological object obligatory} \overset{?}{\Longrightarrow} \perp}$$

Must exist object that is well-defined for Leray-Hopf / suitable weak class, survives viscosity, and derivable from $T^* < \infty$ or $\mathcal{C}_{PROVEN}$.

2 Gate A: Objeto topológico bien definido

Candidates:

- vortex lines: integral curves $\dot{X} = \omega(X,t)$. Requires $\omega \in Lip$ to define uniquely. For suitable weak, $\omega \in L_t^2 L_x^2$ only, not Lipschitz. Lines not well-defined pointwise. For Euler, transport; for NS, diffusion breaks material preservation.
- linking / helicity $H(t) = \int u \cdot \omega$. For Euler H conserved. For NS: $\dot{H} = -2\nu \int \omega \cdot curl\omega$. Not invariant. Not sign-controlled. Bounded by E_0 but $\not\Rightarrow$ regularity. H-01 already showed global invariants coexist with blow-up.
- knots degree/index: needs $\omega \neq 0$ region with non-vanishing and closed loops. Requires ω continuous and non-zero set open. From $\mathcal{C}_{PROVEN}$ only $L^3(B_{r_n}) \geq \varepsilon_0$ for u, no $\omega \neq 0$ open set guaranteed. C_J^{neg} is loss of direction, opposite of regular field for topology.

Well-definedness check vs $\mathcal{C}_{PROVEN}$: - No C^1 vorticity from $C_E^{weak} + C_F^{conc}$. C_I^{freq} $N_c \to \infty$ indicates small-scale creation, making lines less regular, not more. - No open set where $\omega \neq 0$ persists to define degree. Lower concentration on u does NOT imply lower bound on ω topology.

$$\boxed{\text{vortex lines require } \omega \in Lip \text{ or log-Lip - not in } \mathcal{C}_{PROVEN}}$$

$$\boxed{H(t) \text{ not invariant viscous - } \dot{H} = -2\nu \int \omega \cdot curl\omega}$$

$$\boxed{\text{linking/helicity } \not\Rightarrow \text{ obstruction to } T^* < \infty}$$

3 Gate B preview: Evolution under viscous NS

$\partial_t\omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu\Delta\omega$. Stretching term preserves topology materially in Euler, but $\nu\Delta\omega$ causes reconnection, diffusion of lines, breaking of transport. No preserved linking without extra L^∞ control. Would need small ν limit, not available.

4 Gate C preview: Connection $T^* < \infty \Rightarrow P$

Even if topological object defined, need $T^* < \infty \Rightarrow$ object acquires property P (e.g., infinite linking, degree blow-up). No theorem from $\mathcal{C}_{PROVEN}$ gives P. Conserved invariant $\not\Rightarrow$ regularity: $\boxed{\text{invariant} \not\Rightarrow \text{regularity}}$.

5 Gate D preview: $P \Rightarrow \perp$

Would need bridge topology $\Rightarrow$ analytic obstruction to blow-up, not assumed. No known result: infinite vortex linking compatible with smooth solution? For Euler complex topology does not imply blow-up; for NS even less.

6 Gate A verdict

No topological structure of vorticity that is: 1. well-defined for suitable weak class with only $\mathcal{C}_{PROVEN}$, 2. survives $\nu\Delta\omega$ without extra regularity, 3. derivable from $T^* < \infty$.

Candidates either require C^1 vorticity (not available), or are non-invariant viscously (helicity), or coexist with blow-up (H-01).

$$\boxed{\text{G11-02-}GATE\text{-}A = BLOCKED}$$

$$\boxed{\text{G11-02} = CLOSED/BLOCKED \ (\textit{pre-audit})}$$

Consequence: No need to open full B-D machinery for topology with current $\mathcal{C}_{PROVEN}$. Remaining:

$$\boxed{\text{G11-03} : alternative\ ancient\ rigidity\ (last\ frontier)}$$

$$\boxed{G7 \to G8 \to G9 \to G10 \to G11\text{-}01 \to G11\text{-}02 = FROZEN \to CLOSED \to BLOCKED \to CLOSED \to CLO}$$

$$\boxed{NO\ hypothesis\ nueva,\ NO\ reciclaje,\ NO\ paralelo}$$